

Cambridge (CIE) IGCSE Physics
Paper 6: Alternative to Practical (Set D)

Saturday 26 April 2025

Morning (Time: 1 hour 0 minutes)

Total marks

/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) A student is determining the spring constant k of a spring by two methods.

Fig. 1.2 shows how the apparatus is used.

Method 1

On Fig. 1.1, measure the unstretched length l_0 of the spring, in mm.

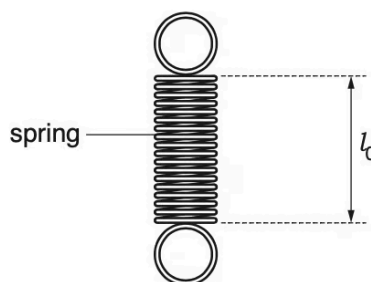


Fig. 1.1

$l_0 = \dots\dots\dots$ mm

(1 mark)

(b) The student attaches the spring to the clamp as shown in Fig. 1.2.

He hangs a 300 g mass on the spring.

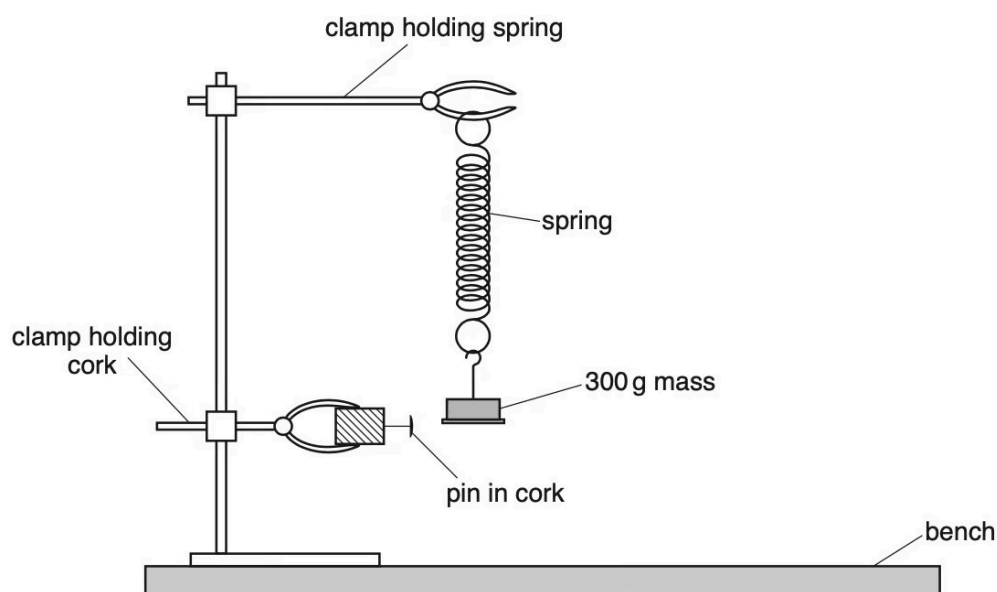


Fig. 1.2

He measures the new length l of the spring.

$$l = 53 \text{ mm}$$

(i) Calculate the extension e of the spring using the equation $e = l - l_0$.

$$e = \dots\dots\dots \text{ mm [1]}$$

(ii) Calculate a value for the spring constant k using the following equation:

$$k = \frac{F}{e},$$

where $F = 3.0 \text{ N}$.

$$k = \dots\dots\dots \text{ N/mm [1]}$$

(2 marks)

(c) Method 2

The student pulls the mass down a short distance and releases it so that it oscillates up and down. Fig. 1.3 shows the time t taken for 10 complete oscillations.



Fig. 1.3

(i) Record the time t taken for 10 complete oscillations.

$t = \dots\dots\dots$ [1]

(ii)

1. Calculate the time T taken for one complete oscillation.

$T = \dots\dots\dots$

2. Calculate T^2 .

$T^2 = \dots\dots\dots$

[2]

(iii)

Calculate the spring constant k using the following equation:

$$k = \frac{0.040m}{T^2},$$

where $m = 0.300\text{kg}$.

$k = \dots\dots\dots \text{N/mm [1]}$

.....

.....

.....

.....

(4 marks)

- (d) State and explain whether your two values for k are the same within the limits of experimental accuracy.

.....

.....

(2 marks)

- (e) A student states that repeating Method 1 with different masses would improve the reliability of the value obtained for k .

Suggest additional values for the mass m that you would use when repeating the experiment to improve the reliability.

.....

.....

(2 marks)

- 2 (a)** A student is investigating how partly covering the surface of the water in a beaker affects the rate at which the water cools.

The apparatus used is shown in Fig. 1.1.

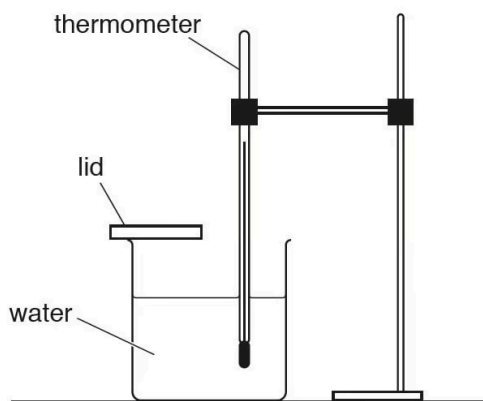


Fig. 1.1

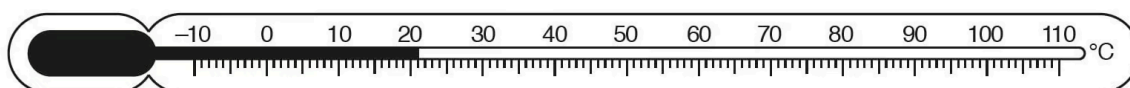


Fig. 1.2

Record the room temperature θ_R , in $^{\circ}\text{C}$, shown on the thermometer in Fig. 1.2.

$\theta_R = \dots\dots\dots$

(1 mark)

- (b)** The student pours 100 cm^3 of hot water into a beaker. She places lid A on the beaker. This leaves half of the water surface uncovered, as shown in Fig. 1.3.

She records the temperature of the water in the beaker and immediately starts a stopclock.

She records the temperature θ of the water every 30 s. Her readings are shown in Table 1.1.

She repeats the procedure using lid B. This leaves a quarter of the water surface uncovered, as shown in Fig. 1.4.

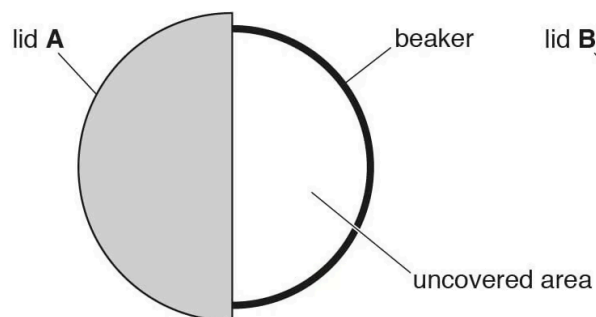


Fig. 1.3

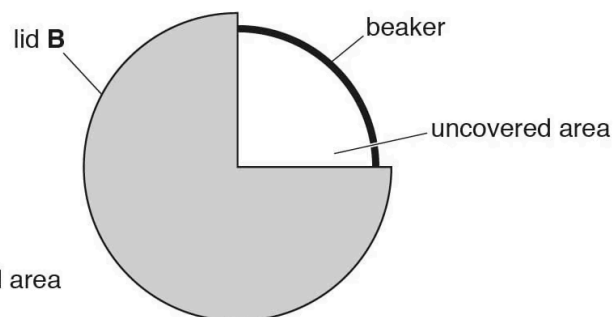


Fig. 1.4

Complete the headings and the time t column in Table 1.1.

Table 1.1

	beaker with lid A	beaker with lid B
$t /$	$\theta /$	$\theta /$
0	80.0	81.0
	77.0	79.0
	74.5	77.5
	72.5	76.0
	70.5	75.0
	69.5	74.0
	68.0	73.5

(2 marks)

- (c) Describe a precaution that should be taken to ensure that the temperature readings are as accurate as possible in the experiment.

(1 mark)

- (d) (i) Write a conclusion to this experiment, stating for which lid the rate of cooling is greater.

Explain your answer by reference to the results.

[2]

- (ii) Suggest a change to the **apparatus** that could produce a greater difference between the rates of cooling for lid **A** and lid **B**.

Explain why the change might produce a greater difference.

[2]

(4 marks)

- (e) A student suggests that the rate of cooling is directly proportional to the percentage of the uncovered surface area. He wants to draw a graph of cooling rate against the percentage of uncovered area to investigate this.

Describe how his graph line will show whether the rate of cooling and the percentage of uncovered surface area are directly proportional.

(2 marks)

- (f) Students in other countries carry out the same experiment.

Suggest **one** factor that they should keep the same if they are to obtain similar readings.

(1 mark)

3 (a) A student is determining the resistance of a piece of wire.

Fig. 2.1 shows the circuit she uses.

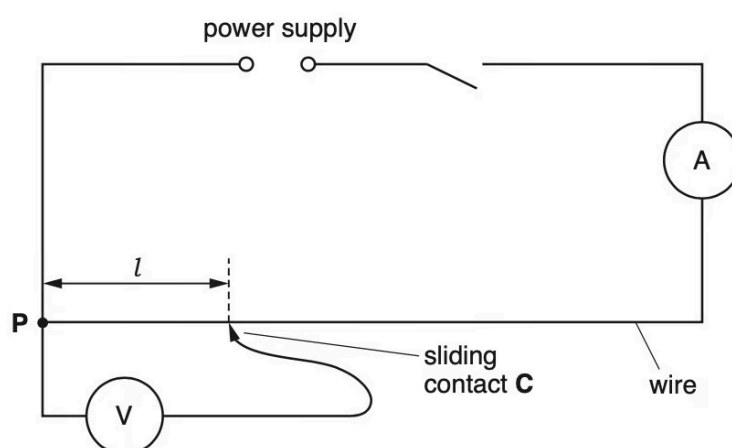


Fig. 2.1

Record the current I in the circuit, as shown on the ammeter in Fig. 2.2.

$I = \dots\dots\dots$

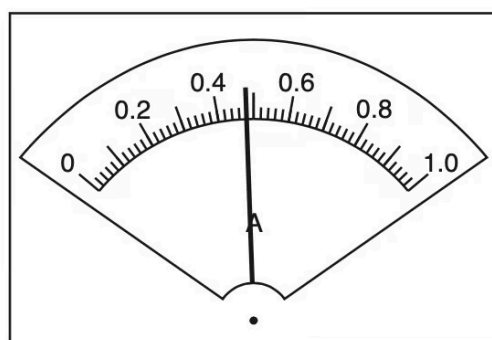


Fig. 2.2

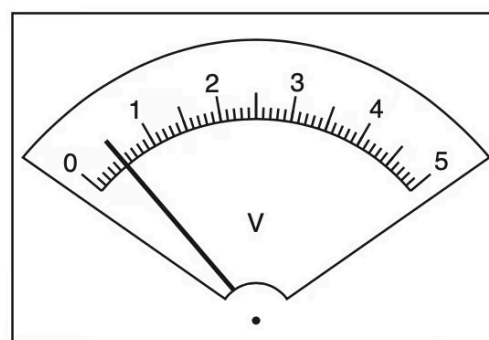


Fig. 2.3

(1 mark)

- (b) The student places the sliding contact **C** at a distance $l = 20.0$ cm from **P**. The voltmeter reading is shown in Fig. 2.3. Record the voltmeter reading in Table 2.1 for $l = 20.0$ cm.

Table 2.1

$l/$	$V/$
20.0	
40.0	0.9
60.0	1.6
80.0	2.0
100.0	2.4

(1 mark)

- (c) The student repeats the procedure using values of $l = 40.0$ cm, 60.0 cm, 80.0 cm and 100.0 cm. The readings are shown in Table 2.1.

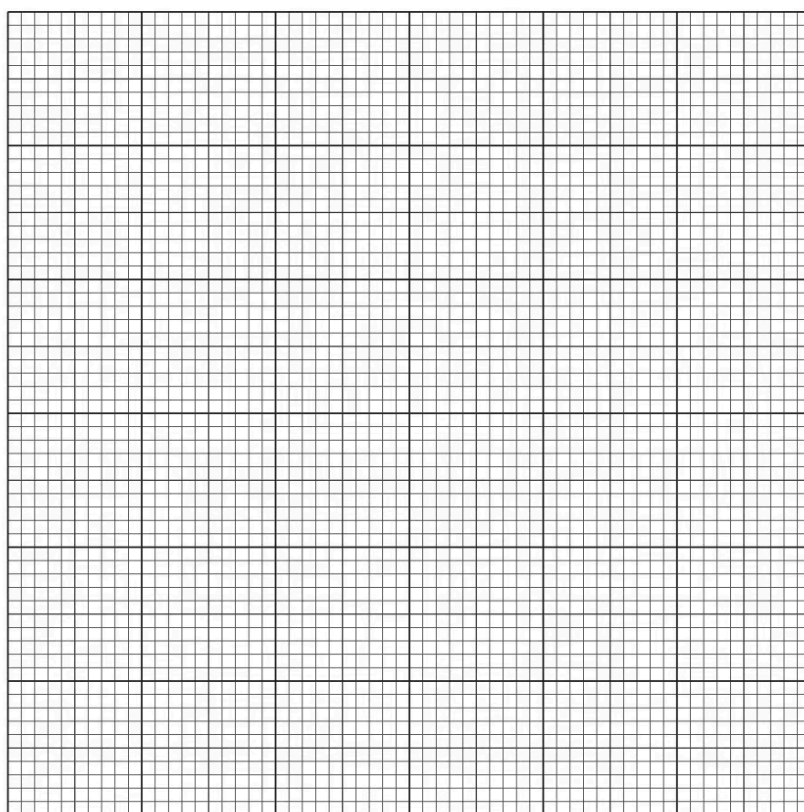
Complete the column headings in the table.

Table 2.1

$I/$	$V/$
20.0	
40.0	0.9
60.0	1.6
80.0	2.0
100.0	2.4

(1 mark)

(d) Plot a graph of V / V (y-axis) against l / cm (x-axis). Start both axes at the origin (0, 0).



.....

.....

.....

.....

(4 marks)

(e) (i) Determine the gradient G of the graph. Show clearly on the graph how you obtained the necessary information.

$G = \dots\dots\dots$ [2]

(ii) Calculate the resistance R of each centimetre of the wire. Use the following equation:

$$R = \frac{Gk}{I},$$

where $k = 1.0 \text{ V/cm}$ and where I is the current recorded in (a).

Include the unit.

$R = \dots\dots\dots$ [2]

.....

.....

.....

.....

(4 marks)

4 A student wants to investigate the effect of air resistance on the swing of a pendulum.

Plan an experiment which will enable him to investigate how air resistance changes the way in which a pendulum swings.

The apparatus available includes:

a light wooden rod, approximately 80cm long with a hole at one end, through which a nail will fit

a piece of modelling clay to act as a pendulum bob, as shown in Fig. 4.1

a sheet of thick card which will provide the air resistance when the pendulum swings.

In your plan, you should:

- list any additional apparatus needed

- explain briefly how you would carry out the experiment including exactly which measurements should be taken
- state the key variables that you would control
- draw a table, or tables, with column headings, to show how you would display your readings (you are **not** required to enter any readings in the table)
- explain how you would use your readings to reach a conclusion.

You may add to Fig. 4.1 or draw an additional diagram if it helps to explain your plan.

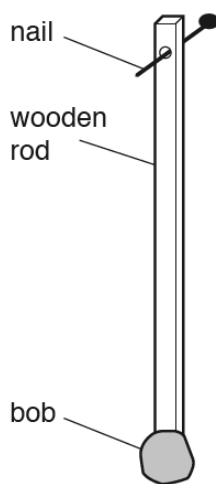


Fig. 4.1

(7 marks)